Team 043 : The Incumbency Paradox in Indian Elections – An Analysis of Win Rates and Margins

The Main Hypothesis/Question

Our project aims to understand whether candidates who have already held office have an advantage in Indian Parliamentary elections, or whether voters tend to favour new candidates over sitting representatives. At first glance, comparing the win rates of incumbents and challengers might seem like a straightforward way to answer this question. However, the results can be misleading because political parties often choose to re-field candidates they believe have a good chance of winning.

To address this issue, we plan to focus on close electoral contests, particularly those in which the winning margin in the previous election was less than 2–5%. By studying these closely contested seats, we hope to reduce the effect of candidate selection and get a clearer picture of the personal advantage, if any, that comes with incumbency. This approach is inspired by regression discontinuity methods, which are commonly used to study the effects of narrowly winning an election.

Source Papers, Blogs, and Podcasts

We will use existing research on electoral outcomes and incumbency effects to guide our analysis and compare our findings with previous studies.

- Linden, L. (2004). *Are Incumbents Really Advantaged? The Preference for Non-Incumbents in Indian Elections.*
- Uppal, Y. (2009). *The Disadvantaged Incumbents: Estimating Incumbency Effects in Indian State Legislatures.*
- Trivedi Centre for Political Data (TCPD) Codebooks and release notes, to understand the dataset and the variables used.

Programming Languages or Tools/Software Intended

We plan to use Python for data cleaning, statistical analysis, and visualisation. The main tools we intend to use are:

- **Programming Language:** Python 3.
- **Libraries:** `pandas` for organising and cleaning the data; `scipy.stats` for binomial tests, chi-square tests, t-tests, and related statistical analysis; `statsmodels` for confidence intervals and regression models; and `matplotlib`/`seaborn` for visualising election margins and distributions.
- **Document Preparation:** LaTeX for preparing the final report and generating the PDF.

Datasets

- **Primary Dataset:** We will use the TCPD Lok Dhaba Dataset (TCPD-IED), which covers Indian elections from 1962 to 2024. The data is available through lokdhaba.ashoka.edu.in and is also mirrored on Data OpenCity under the CC BY-NC-SA 4.0 licence. It includes person IDs that can help us track candidates across elections, along with variables such as `No_Terms`, `Margin_Percentage`, and `Turnout_Percentage`. These will be useful for studying repeat candidates, party switching, electoral margins, and voter turnout.

  Link to the dataset

- **Secondary Dataset for Verification:** We will refer to the Election Commission of India (ECI) Statistical Reports and India Data Portal wherever necessary to cross-check the data and verify important election results.

Logistics

- **Updated Spreadsheet:** No updates at the moment.
- **Proposed Division of Work:** We have divided the work among team members so that each person can focus on a specific part of the project.
  - **25b0963:** Collecting and combining the required datasets, cleaning the data, and checking consistency. This will also involve accounting for constituency boundary changes using `DelimID`.
  - **25b0981:** Preparing the data for analysis by identifying close races, tracking candidates across elections, and creating a variable to identify candidates who switch parties (turncoats).
  - **25b1037:** Carrying out the initial statistical analysis, including binomial and chi-square tests, and estimating the parameters of a geometric distribution to study consecutive electoral wins.
  - **25b1079:** Analysing election margins through Q-Q plots and normality checks, examining covariance and correlation, and building a multiple regression model to study the relationship between election margins, voter turnout, and the effective number of electoral parties (ENOP).
- **Collaboration Plan:** We will use GitHub to store our code, track changes, and collaborate on the project. This will allow all team members to contribute to the analysis while keeping the work organised and easy to review.